From Energy to Air: A Predictive Model of State Air Quality

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Introduction

Our project investigates the relationship between energy production, energy consumption, and air quality across the United States from 1980 to 2024. Using data from the U.S. Energy Information Administration's State Energy Data System (SEDS) and the Environmental Protection Agency's Air Quality System (AQS), we aimed to identify patterns in states' energy profiles and determine whether those profiles can help explain or predict differences in air quality.

This analysis will help us better understand the relationship between energy use and air quality. Additionally, the findings could help policymakers, researchers, and environmental planners evaluate how different energy profiles relate to environmental outcomes.

Our team's motivation for this project stemmed from one team member's prior experience with the AQS dataset (Michael's Milestone 1 project; see example 4 in related work). However, we wanted to pursue a new cross-analysis and found the SEDS dataset. After some initial analysis, we decided as a team that there could be some interesting findings in combining AQS and SEDS data for a more rounded look at explaining air quality changes.

For supervised learning, we trained and compared multiple regression models, including linear regression with regularization, random forest, k-nearest neighbors (KNN), and support vector regression (SVR), to predict state-level Air Quality Index (AQI) using energy features from the U.S. Energy Information Administration's State Energy Data System (SEDS). The training data spans 1980-2019, and the test data spans 2020-24. We used the forward-chaining cross-validation and hyperparameter tuning methods to evaluate model performance while reducing the risk of data leakage. For unsupervised learning, we applied K-means and agglomerative hierarchical clustering to group states according to similarities in their energy production and consumption profiles from 1960 to 2024. Additionally, we compared the resulting clusters using separation metrics, stability measures, visualizations, and sensitivity analysis.

This project contributes to existing research by combining historical energy data and air quality measurements with both predictive modeling and energy-profile clustering. Unlike studies that often use physics-based simulations to project future pollution levels, our approach uses historical observations to assess whether energy patterns explain or predict air-quality differences.

Our supervised learning results showed that the random forest model achieved the best predictive performance among the models evaluated, with a mean cross-validation RMSE of 3.78 AQI points using energy and state features. However, a model using only state and year performed similarly, with a mean RMSE of approximately 3.68. This suggests the energy features added little predictive value.

Our unsupervised learning results showed that both K-means and agglomerative hierarchical clustering identified groups of states with similar historical energy production and consumption profiles. Using seven

clusters, hierarchical clustering achieved a silhouette score of 0.155 and an adjusted rand index (ARI) stability score of 0.650, compared with 0.149 and 0.621, respectively, for K-means. We used hierarchical clustering for the final model because it had a slightly higher silhouette score, better stability, and greater overall reproducibility than K-means.

Related Work

Example 1: State-level drivers of future fine particulate matter mortality in the United States. This study examines the relationship between states' energy and fuel choices and air quality, specifically PM2.5 (particulate matter 2.5 microns in diameter or less), through a human-earth model. Our project takes a different approach by using historical SEDS and AQS data to train machine learning models, including regression and random forest. Rather than relying on a physics-based simulation to predict conditions through 2050, our analysis uses real-world historical data to identify patterns and relationships.

Example 2: Regional air pollution mixtures across the continental US. This study used K-means clustering to group air-quality monitoring sites across the United States based on PM2.5 chemical composition from 2008 to 2016. We identified 27 clusters with distinct pollution profiles and geographic patterns, demonstrating a method that helps identify regional differences in environmental conditions. While they clustered monitoring sites by pollution composition, our team clustered states by energy production and consumption profiles and separately tested whether they could help predict AQI.

Example 3: Predicting PM2.5 Concentrations Across USA Using Machine Learning: This study used several different models to predict PM2.5 concentrations using air-quality monitoring data from 2017 to 2021. Their results showed that random forest and support vector regression performed best overall, although prediction accuracy varied depending on location. In contrast, our study examines whether historical energy production and consumption can predict air quality, rather than relying solely on air-quality monitoring data.

Example 4: Winter 2025 Milestone 1 Project. Michael's project analyzed EPA AQS air-quality data from 2016 to 2023, along with the Department of Energy's (DoE) fuel station and vehicle registration data. The goal was to see if EV adoption influenced air-quality metrics.

Data Sources

State Energy Data System (SEDS)

The U.S. Energy Information Administration (EIA) State Energy Data System (SEDS) provides historical energy estimates for U.S. states. This analysis utilizes Excel files spanning 1960 to 2024 for all production estimates and all consumption estimates in British thermal units (Btu).

To download the data from the website, go to:
Data Files → Production → All production estimates → XLSX
Data Files → Consumption → All consumption estimates → in BTU → XLSX

Because these files organize annual data by state and 5-character Master System of Names (MSN) codes, we created a custom categories spreadsheet using the EIA guide to map codes to readable descriptions and filter variables.

Key variables (with guidance from EIA documentation) include state, year, population, and energy production and consumption by source and sector. Energy sources encompass fossil fuels (coal, petroleum, natural gas), nuclear energy, and renewables, eventually split into combustible (biofuels, wood/waste) and non-combustible (hydro, wind, solar, geothermal) types. We used population (TPOPP) to calculate per-capita energy values but excluded it as a direct clustering feature to prevent state size from dominating the results.

Air Quality System (AQS)

We utilized the Air Quality System (AQS) data by county and year from the United States' Environmental Protection Agency (EPA). The AQS data has annual air quality metrics reported by county. Important variables include state, county, year, total days with AQI data, days with PM2.5 data, days with ozone data, and Median AQI in that county. AQS data is stored as separate zip files of CSV files per year. The data spans from 1980 to 2024.

To download the data from the website, go to: Annual Summary Data → AQI by county → Download each zip file.

Exploratory Data Analysis

The AQI EDA looked at air quality data across different states, counties, and years to identify data-quality issues, coverage gaps, changes in AQI over time, and relationships between pollutants. The analysis found that air quality data was not consistently available across all states and years, with the AQI heatmap revealing some states having more complete data than others.

In contrast, the SEDS EDA looked at energy production and consumption across 51 geographic entities from 1960 to 2024 to identify trends over time, state differences, variable distributions, and relationships among energy sources. The analysis found differences in energy production patterns across states over time, and the correlation matrix showed strong positive relationships among certain variables, notably those related to energy production.

See Appendix A for the AQI heatmap and the SEDS correlation matrix.

Feature Engineering

SEDS Cleaning

We took the steps below to go from two raw spreadsheets of data and a supporting category spreadsheet to one dataset ready for modeling. Supervised learning and unsupervised learning utilized largely the same steps; however, supervised learning did not average into windows due to needing to predict yearly data. Additionally, one-hot encoding was only performed for supervised learning because state was not included as a variable for unsupervised learning to avoid data leakage. Supervised learning also fit steps 5, 6, 9, 10, and 11 below on the training data, while unsupervised learning fit these on all data.

1. Combine production and consumption data: two files into one for simplicity

2. Filter for variables from categories dynamically: provides flexibility for future variations
3. Drop non-US states aside from DC (such as country aggregation and territories)
4. Transpose from year columns to energy variable columns while keeping states as rows: to allow for time-based analysis
5. Drop duplicated features: minimize double counted feature effect
6. Drop nearly empty variables: nearly empty data will create outliers when data is present
7. Combine non-combustible renewables: non-combustible renewables behave the same during an air quality analysis, as they have significantly less emissions than combustible renewables or fossil fuels
8. Transform to energy trajectories: averaging into windows smooths noise while keeping time trends
9. Transform to per capita: removes effect of state size so that larger states like TX or CA do not dominate smaller states like WY
10. Transform with log1p: minimizes skewing of data while accounting for zero and negative values (both of which are possible and exist in the data)
11. Transform with standard scaling: puts energy variables on the same scale
12. One-hot encoding of categorical data (state and cluster id): stops the model from assuming an ordinal relationship between variables

To view the final list of features, see Appendix B.

AQS Cleaning

The steps below transformed the AQS data into one CSV for supervised learning.

1. Unzip CSVs and merge into one dataframe
2. Drop non-US states aside from DC and align state names with the SEDS dataset
3. Aggregate county data to state-level to align with SEDS
    a. Drop county-years with under 180 days of data. These rows had higher variance.
    b. Drop county-years with no ozone or PM2.5 monitoring. These rows were not representative.
    c. Compute a days-weighted average median AQI for each state-year.

To view the final list of features, see Appendix C.

Supervised Learning

Methods Description

Our project's goal was to predict air quality in a state for a particular year based on its energy production and consumption features by type and sector. We utilized the aggregated weighted median AQI for each state as the target feature. In addition, we used the normalized and scaled SEDS data for the predictive features (see data sources and feature engineering sections). We added the clusters from unsupervised learning as a predictive feature to see if these clusters improved predictions.

We used the sci-kit learn implementations of four model families:

1. Linear regression: captures linear relationships between input variables and the target. We utilized four different types of linear regression: ordinary least squares (OLS), ridge regression, lasso regression, and ElasticNet regression. We implemented an OLS model as a baseline and the other models to introduce regularization. Several features were highly multicollinear, so we trained regularized models to reduce overfitting.
2. Tree ensemble regression: captures non-linearities. We utilized random forest regression and histogram gradient boosting tree regression.
3. K-nearest neighbors (KNN) regression: makes predictions based on similar instances and can capture non-linear behavior as well.
4. Support vector regression with an RBF kernel: utilizes built-in regularization and can capture non-linear relationships.

We used five forward-chaining folds over 1995-2019 to reduce leakage when training the model. Utilizing a shuffled K-fold instead would have trained the model on future and past years when predicting a state-year, which would have introduced leakage. We tuned hyperparameters using GridSearchCV on forward chaining folds and evaluated using RMSE to select the best model.

Supervised Evaluation

Overall Results

We utilized root mean standard error (RMSE) and mean absolute error (MAE) as our primary evaluation metrics. RMSE and MAE are in our target's units, so they are relatively interpretable. We chose RMSE as our primary evaluation metric because it minimizes large errors.

To find the most effective feature set, we trained models on several different feature sets: only SEDS energy features, SEDS energy features with clusters, energy and state, and state and year without energy.

The random forest tree ensemble model trained on energy and state features had the lowest average RMSE (3.78 ± 0.56), 54% lower than the baseline OLS regression model (8.28 ± 6.30).

The best models with only state and year data and no energy features performed similarly for AQI predictions, showing a potentially limited impact of the energy features. The model without energy features had a slightly lower mean RMSE than the best model trained on energy and state (3.68 ± 0.60 vs. 3.78 ± 0.56); however, it was within the standard deviation of both.

Model family	**Energy**	**Energy + cluster**	**Energy + state**	**state+year (no energy)**
Instance-based (KNN)	4.56 ± 1.20	4.58 ± 1.19	4.57 ± 1.18	3.68 ± 0.60
Kernel (SVR)	4.97 ± 1.03	4.99 ± 1.08	4.95 ± 1.10	4.38 ± 1.00
Linear (Ridge/Lasso/ElasticNet)	5.68 ± 1.04	5.66 ± 0.98	5.68 ± 1.04	4.98 ± 0.77
Tree ensemble (RF/HGB)	3.84 ± 0.54	3.83 ± 0.58	3.78 ± 0.56	3.71 ± 0.63

Figure 1: CV RMSE across the best model in each family ± std dev across five forward folds

Cluster IDs did not appreciably impact predictions, and the feature was not included in the best model. The best random forest model's RMSE was 3.78 ± 0.56 with no clusters. It was 3.83 ± 0.58 with the clustering variable included. The clusters included 1960-2024 SEDS data. Since it includes 2020-2024 test data, it could have introduced minor feature leakage if it was included in the final model.

Feature Importance and Ablation Analysis

To determine feature importance, we utilized grouped permutation importance and SHAP. Permutation importance shows how much the best model depends on a group of features. It is measured in the units of the target, so it is easy to interpret. Since feature groups such as energy source and sector were highly correlated, we combined them. Permutation importance identified three energy-source feature groups with the largest impact: natural gas (+1.19), petroleum (+0.91), and renewables (+0.90).

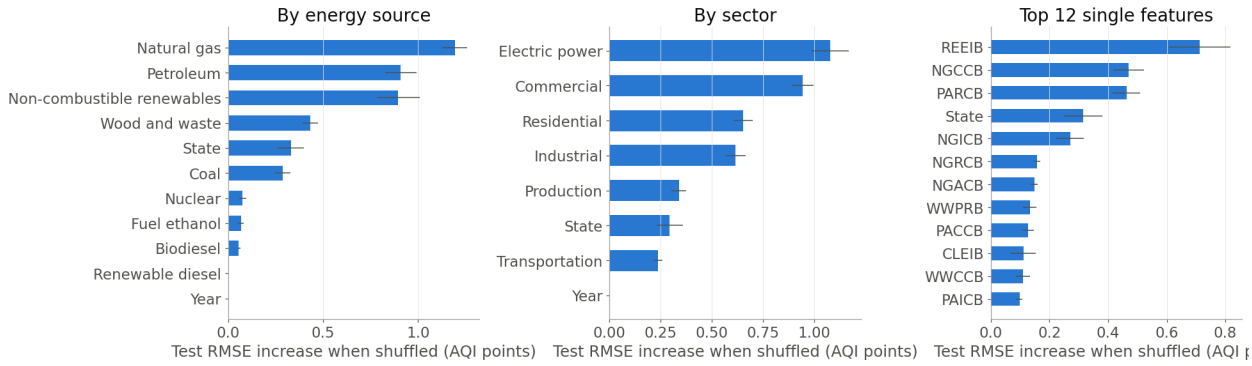


Figure 2: Permutation importance of feature groups for random forest model

We also used SHAP to analyze the impact of the year feature on the model. The permutation importance of the year feature was zero but because SHAP shows how the feature affects individual predictions, the SHAP analysis showed that year does impact predictions. According to the SHAP analysis, the impact of year varies over time and is smaller on the test data. The training absolute average value was 0.72, and the test absolute average value was 0.22.

Ablation analysis showed that no single feature group had a large impact on model performance when removed during training. Coal (+0.07) was the only feature that made performance worse across all five folds when removed.

Sensitivity Analysis

We retrained the best model with several different hyperparameters used during CV training (minimum lead size, depth, and features per split) to determine the impact of the model's sensitivity to these hyperparameters. The model was sensitive to the minimum leaf size hyperparameter (0.7 RMSE increase) but not to max depth (0.10 RMSE increase) and max features per split (0.04 RMSE increase).

We also examined the model's sensitivity to the duplicate threshold for dropping features during pre-processing. We found little impact from reducing the threshold from 0.98 to 0.9 (3.78 ± 0.56 vs. 3.76 ± 0.56 RMSE).

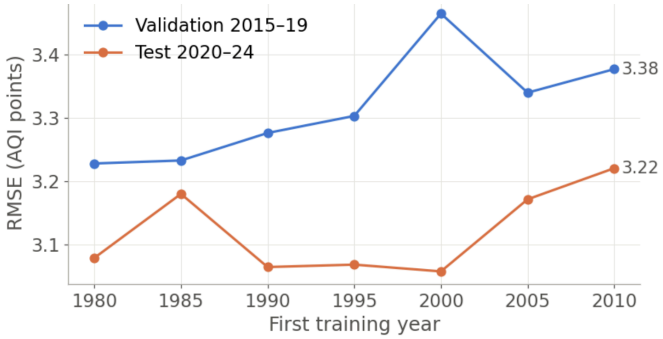


Figure 3: Training-window sensitivity of the random forest model

Finally, we analyzed the model's sensitivity to the training window. The longer window includes more data, but this data may be outdated, such as due to different emission standards. To examine this, we re-trained the model with data starting at later years (1980, 1985, etc.). We found that the model was sensitive to the training window, especially after 2000. Test RMSE was 3.08 with the full AQI training dataset (1980-2019) but 3.22 RMSE on a training dataset only including 2010-2019 data.

Tradeoffs

There was a tradeoff between accuracy and interpretability when building the model. The random forest model has the lowest RMSE but is less interpretable than linear models such as Lasso, since Lasso models have limited features and show directions of impact from each feature.

Accuracy vs. speed is another tradeoff. The random forest model with the lowest RMSE took about 1,000 times longer to fit than the linear models or KNN. This only added four seconds on the device used to train the model, so it should not drive the choice of model, but it may add additional time for devices with less computational power.

Failure Analysis

Prediction failed for several states, including Hawaii, Nebraska, and Rhode Island. Potential reasons include missing variables, changes in AQI measuring systems, and variation due to a limited number of counties reporting data. The chart below shows predicted AQI (blue), actual AI (grey), and number of counties with data (orange) during the time period for those three states. It shows the direction and magnitude of errors and how they corresponded (or did not correspond) to the number of counties in the state that affect the weighted AQI.

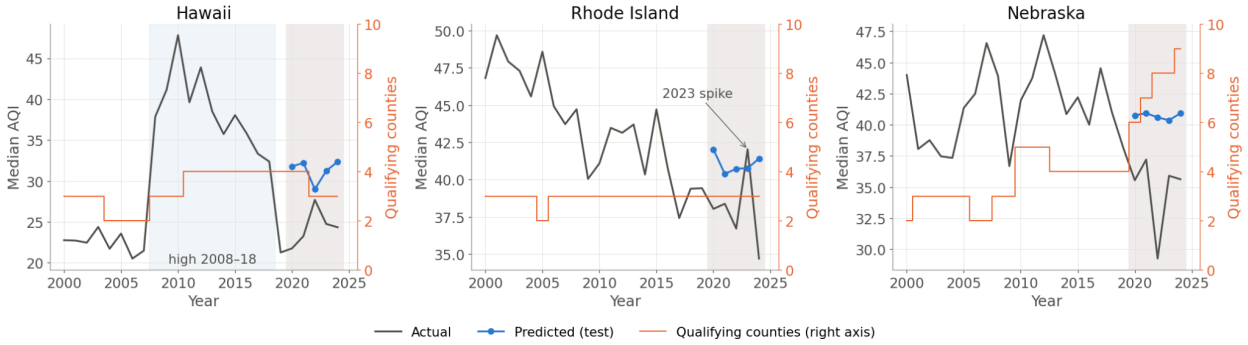


Figure 4: Failure Analysis by State: Median predicted and actual AQI and number of counties vs year

Missing variables: Volcanic smoke. Hawaii's predicted values were too high for each test year (2020-24). There was a volcanic eruption from 2008-2018 at Mount Kilauea (USGS), which could have impacted air quality, creating a higher baseline in training data than test data. Adding volcanic and weather data would resolve this issue. Wildfire smoke may affect other records as well; Rhode Island's AQI spike in 2023 corresponded to significant Canadian wildfires (Yu).

Changes in measurement systems: Additional AQI measurements in a state. Changes in counties that qualified for their AQI values also may have adversely affected predictions for Nebraska. More counties in Nebraska reported more days with AQI during the test period than the training, raising the number of counties where this data was aggregated from five or less before 2020 to nine in 2024. Without these additional counties, the AQI data for Nebraska would have been much closer to test data. The filtering thresholds for removing county-years influenced this and could be changed in future analysis to mitigate this issue.

Limited monitoring: Few counties. Limited monitoring also potentially influenced the air quality predictions for Rhode Island. Rhode Island only has three monitored counties. There was a sudden fluctuation in the state's AQI in 2023; this variation may have been exacerbated by the limited number of counties, an issue that is mitigated in larger states. This is an artifact of the data structure and the weighted average used to calculate median AQI. A county-level analysis could resolve this (see discussion).

Unsupervised Learning

Methods Description

The objective of this portion of the project was to group the 50 U.S. states and Washington, DC, into clusters based on their energy production and consumption profiles using the SEDS data. We used data covering 1960 to 2024. Since the states did not have predefined energy-profile labels, unsupervised learning was appropriate for identifying similarities and differences without requiring a target variable.

We explored two clustering methods: agglomerative hierarchical clustering and K-means. We utilized the same feature representation for both so their results could be compared fairly.

Agglomerative Hierarchical Clustering

We selected agglomerative hierarchical clustering because it provides a hierarchy of similarities between states. To determine whether the resulting groups depended on the method used to define cluster similarity, we compared four linkage methods (Ward, complete, average, and single) across candidate values of k. This comparison helped determine whether the resulting groups depended on the method used to define cluster similarity.

Hyperparameter exploration compared candidate cluster counts and model configurations using internal evaluation metrics, cluster sizes, and stability. The best-performing configuration from each method family was then examined through visualizations and sensitivity analysis. This approach was important because a single metric cannot fully describe whether clusters are both mathematically separated and useful for interpreting state energy profiles.

K-Means

K-means clustering was selected because it is a common, computationally efficient baseline that produces interpretable groups based on energy-profile similarity (and a different clustering process from hierarchical clustering). The number of clusters (k) and initialization method were explored. Both k-means++ and random initialization were evaluated, with multiple runs and restarts used to reduce the effect of random starting positions and local optima.

Unsupervised Evaluation

Overall Results

Three internal clustering metrics for separation were used. The silhouette score measures how close each state is to its assigned cluster compared with the nearest alternative cluster; higher values generally indicate better separation. The Calinski–Harabasz index compares between-cluster dispersion with within-cluster dispersion, where higher values are generally preferred. The Davies–Bouldin index measures similarity between clusters, with lower values preferred. These metrics were considered together because no ground-truth labels were available and each metric captures a different aspect of cluster quality (Collins-Thompson, 2025).

However, three additional measures were included to evaluate whether the clusters were reliable and practically interpretable, rather than relying only on separation metrics. Stability measures how consistently the overall state groupings are reproduced across repeated clustering runs, with higher values indicating greater agreement. This is measured with the adjusted rand index (ARI). An ARI of 1 indicates identical groupings, while values near 0 indicate little agreement beyond chance. ARI was useful here because silhouette scores calculated on different feature representations are not directly comparable. The least-stable cluster (Jaccard) focuses on the cluster that is reproduced least consistently, helping identify whether a model contains an unreliable group that an overall stability score might hide. Lastly, the number of clusters containing only one or two states was tracked to identify very small clusters. Although small clusters can represent genuinely unusual state energy profiles, they can also indicate that the model is isolating individual states rather than finding broader, interpretable patterns. Together, these measures complement the three internal metrics by assessing reproducibility and cluster usefulness, not just separation.

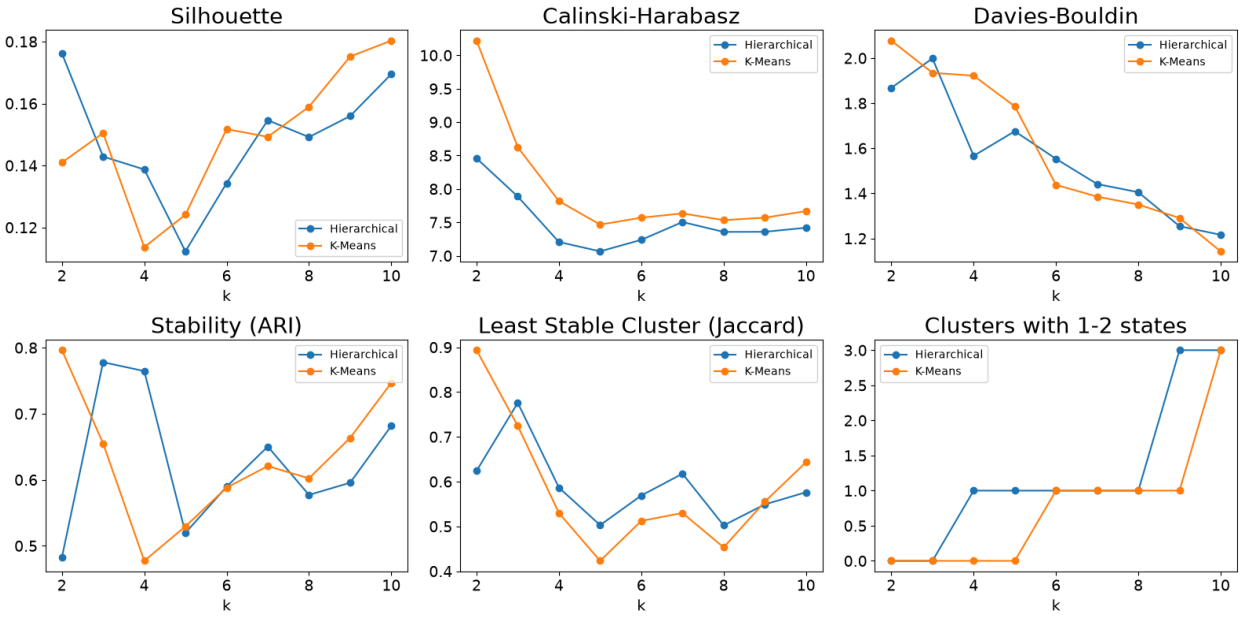


Figure 5: Evaluation Metrics Across Different K

The selected configurations both used seven clusters. K-means used k-means++ initialization, while hierarchical clustering used Ward linkage.

Metric (k=7)	Hierarchical (Ward)	K-Means
Silhouette Score	0.155	0.149
Calinski-Harabasz Index	7.504	7.633
Davies-Bouldin Index	1.441	1.385
Stability (ARI)	0.65	0.621
Least-Stable Cluster (Jaccard)	0.618	0.53

Figure 6: Evaluation Metrics for Best K

The results show that neither method clearly outperformed the other on every metric. K-means had slightly better Calinski–Harabasz and Davies–Bouldin scores, while hierarchical clustering had a slightly higher silhouette score and better stability scores. Both silhouette values were relatively low, suggesting that the state energy profiles did not form completely distinct groups. Some states likely have characteristics that overlap multiple clusters.

Hierarchical clustering was selected as the final model because its overall separation was comparable to K-means, its least-stable cluster had a higher Jaccard score, and its ability for a dendrogram provided a useful way to examine relationships between states. The choice therefore reflects interpretability and stability rather than a claim that hierarchical clustering was best in every respect.

Four complementary visualizations supported the comparison. Silhouette plots showed the distribution of state-level separation scores within each cluster, helping identify weakly separated groups and borderline states.

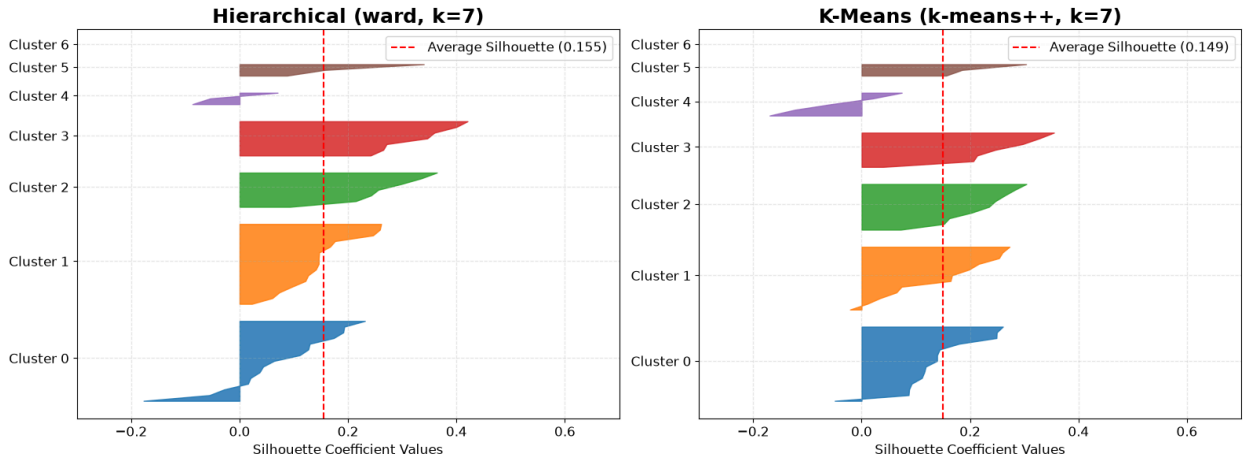


Figure 7: Silhouette Plots

PCA scatter plots projected the high-dimensional feature vectors into two dimensions to visualize the overall grouping structure, although this projection cannot preserve every relationship in the original feature space.

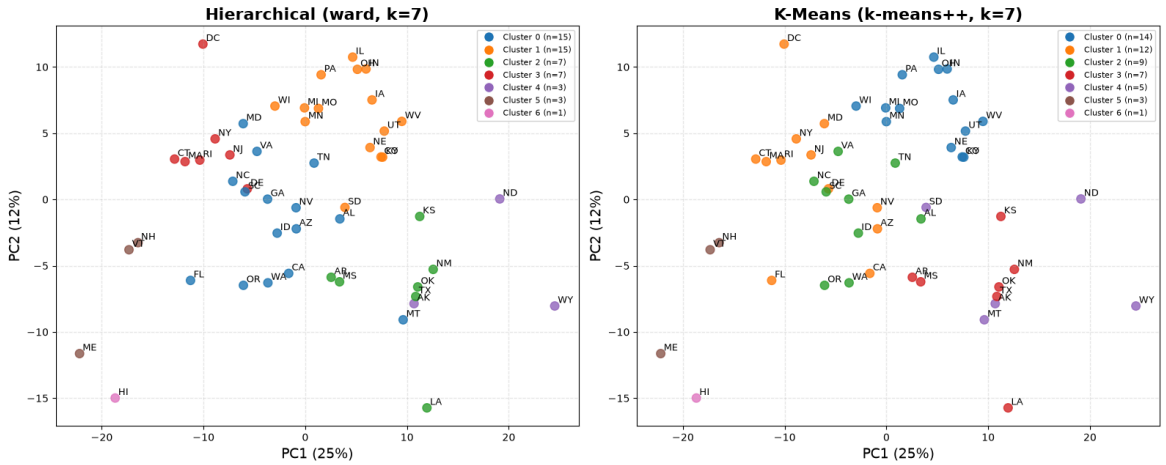


Figure 8: PCA Scatter Plots

Choropleths showed the geographic distribution of clusters and helped reveal regional patterns.

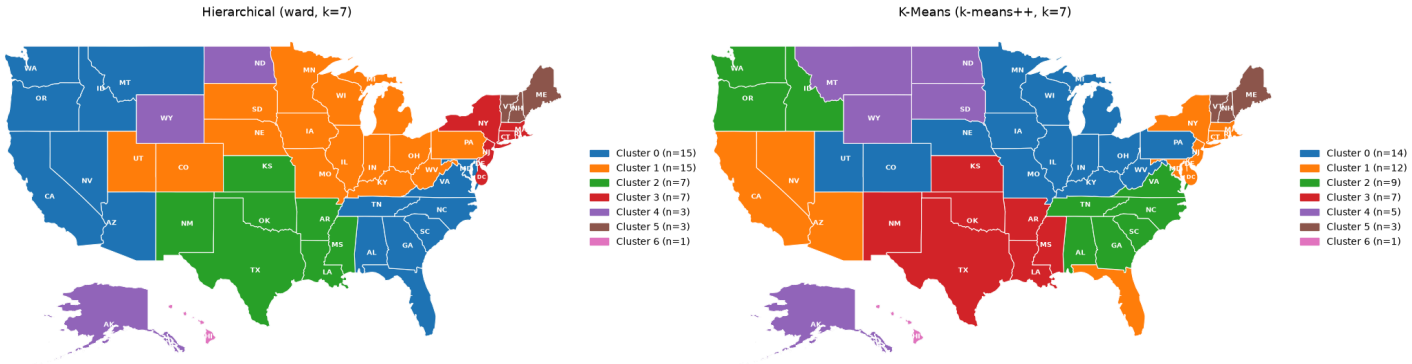


Figure 9: Choropleth Maps

Finally, the hierarchical dendrogram illustrated which states and groups merged at different distances, providing additional insight into the hierarchical model.

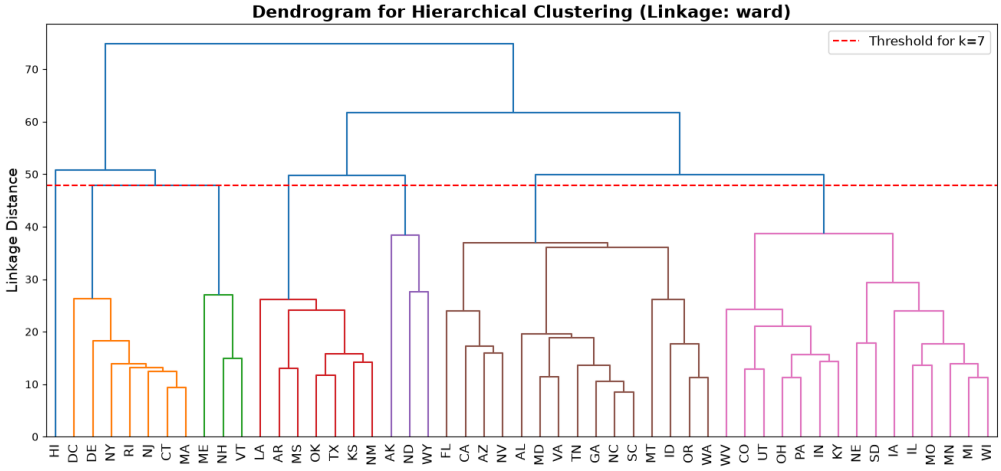


Figure 10: Dendrogram Plot

These plots complement the numerical metrics rather than replacing them.

Sensitivity Analysis

Sensitivity analysis was conducted on the final Ward hierarchical model with seven clusters. Each experiment changed one feature-engineering decision, re-ran the clustering, and compared the resulting assignments with the baseline using ARI.

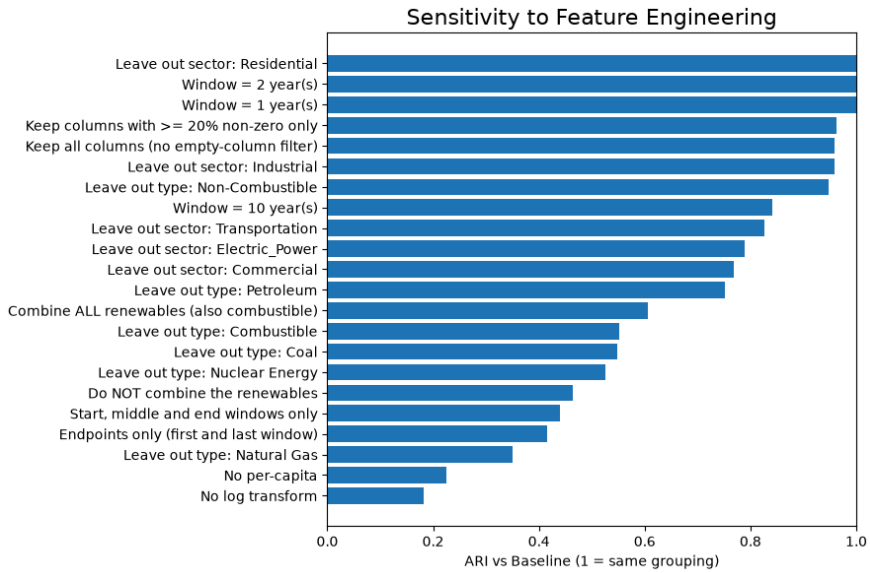


Figure 11: Sensitivity Analysis

The analysis showed that some preprocessing decisions substantially affected the results. Removing the per-capita transformation reduced ARI to 0.225, while removing the log transformation reduced it to 0.182. These findings suggest that accounting for population and reducing the influence of extreme energy values were important in determining the state groupings.

Feature construction also affected the results. Combining all renewables, including combustible sources, produced an ARI of 0.606, while not combining non-combustible renewables reduced ARI to 0.465. This indicates that the treatment of energy categories changes the resulting clusters and supports retaining the distinction between combustible and non-combustible renewables. In contrast, removing columns with fewer than 20% non-zero values resulted in an ARI of 0.963, indicating that this particular filtering choice had relatively little effect on the overall assignments.

The time window also influenced the results. Using ten-year rather than five-year windows produced an ARI of 0.841, whereas one-year and two-year windows preserved the baseline assignments in this experiment. The findings suggest that the model is reasonably robust to some feature-selection choices but more sensitive to transformations and the representation of energy use over time.

Overall, the sensitivity analysis reinforces that clustering outcomes depend on how the data are represented, not just on the algorithm selected. The final clusters should therefore be interpreted as

useful summaries of similarities in historical state energy profiles rather than definitive or universally distinct categories.

Discussion Part A

We were surprised that the predictive power of the SEDS variable model was roughly the same as the state-and-year model. We expected the SEDS features to have more predictive power. Not finding significant predictive power is in itself telling that there are other important factors in determining AQI.

One challenge we encountered was the misalignment between AQI and SEDS data and the need to aggregate AQI data. We ultimately weighted by county and year and removed counties with limited data. Air quality measurement is inconsistent in many counties and would be biased toward big states with many counties, so we decided to keep it at the state level.

We also encountered issues during cross-validation. Initially, we used a shuffled k-fold cross-validation. The models appeared to be severely overfitting, even with very high regularization in linear regression. It was not training on all states each fold and was using future data to predict past data, creating a model that significantly underfit the data. Using the forward-chaining CV resolved this.

With more time and resources, it would be helpful to add data about confounding variables, such as weather and fire, when predicting AQI. This would allow for a more robust and complete model. We would also like to extend this solution to examine county-level air quality trends as well. This report's findings were limited due to the state-aggregation, and analyzing by county could restore additional granularity, although it would likely need additional county-level features to explain many of the differences between counties.

Discussion Part B

One surprising result was how difficult it was to form clearly separated clusters using data from 1960 to 2024. Low silhouette scores for K-means (0.149) and hierarchical clustering (0.155) suggest that states share overlapping energy profiles rather than distinct groups. While both methods produced similar overall results, the sensitivity analysis highlighted how heavily preprocessing choices like population normalization and log transformation influence cluster assignments.

A major challenge was preparing a dataset across many energy types, sectors, and years while handling data sparsity and zero values in earlier periods. This was addressed by combining renewable sources and aggregating into five-year windows. To ensure meaningful results, we evaluated multiple metrics, compared two clustering methods, and ran sensitivity tests.

With more time and resources, we would investigate additional feature representations and explore dimensionality reduction, like PCA, for modeling rather than just visualization. Another extension would be to analyze smaller or less stable clusters to see whether they represent genuinely distinct profiles or are sensitive to modeling choices.

Ethical Considerations

One ethical issue with our supervised learning models is public health risk. Incorrect predictions could lead officials to underestimate pollution levels and overlook health risks to nearby communities, especially

those without the resources to withstand environmental issues. One way to address this is to have domain experts review the model and compare its findings with reliable air-quality data.

Another ethical issue with the supervised learning model is potential bias. States or counties with more observations may cause the model to learn patterns reflecting those areas. Data science models can favor the majority population while overlooking smaller groups (Sandvik). In this project, that means overlooking smaller states or communities with distinctive energy profiles or environmental challenges. One way to combat this is treating the model as a decision-support tool rather than the sole source of information.

One ethical issue to consider with the unsupervised learning model is perpetuating bias based on previous data. There is a possibility that the clusters could be misinterpreted as definitive classifications of states rather than patterns identified by a particular model. For example, the silhouette scores for K-means (0.149) and hierarchical clustering (0.155) were relatively low, indicating a substantial overlap between state energy profiles. Data science relies on historical evidence and can reproduce past patterns without considering the conditions that originally shaped them (Sandvik).

In this case, historical energy patterns might reflect past infrastructure investments, economic conditions, and energy policies. However, they might not indicate what energy policies would be most appropriate for future use. To address these concerns, all findings should be presented with appropriate limitations, and all cluster assignments should not be treated as definitive.

Statement of Work

Michael located, verified, and cleaned the SEDS data and led the unsupervised learning portion of the project. Hayley led the supervised learning section and located, verified, and cleaned the AQI Data. Sravan assisted with data cleaning and led the exploratory data analysis for the project, as well as taking a lead role in writing the proposal, verifying all references, and researching examples of existing projects or studies related to this team's project. The team utilized generative AI models during coding portions of the project (see references for specific models used). All outputs were verified through human-in-the-loop validation.

Project repository can be found here: https://github.com/micleone/SIADS_696_F26_Team10/tree/main.

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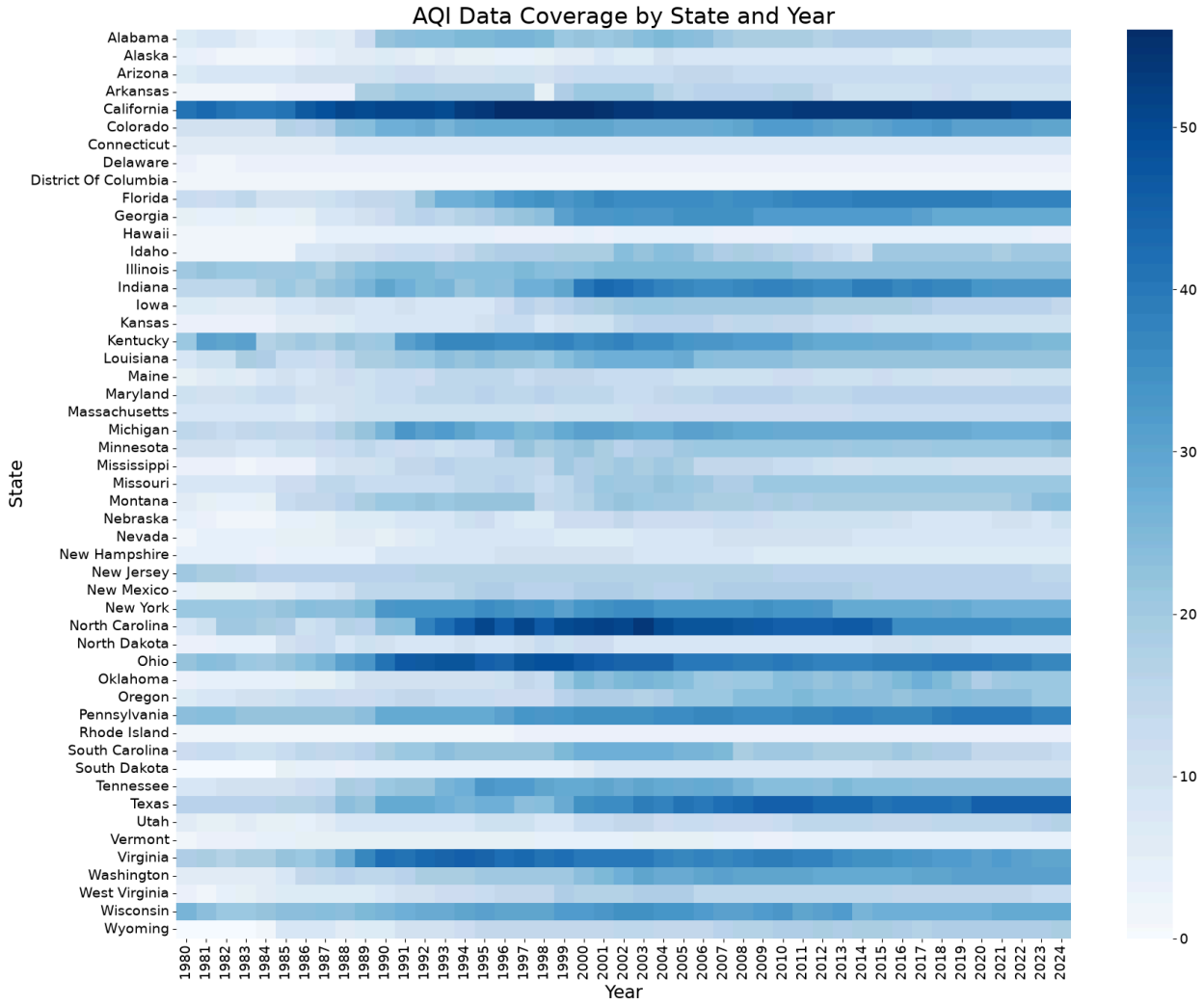
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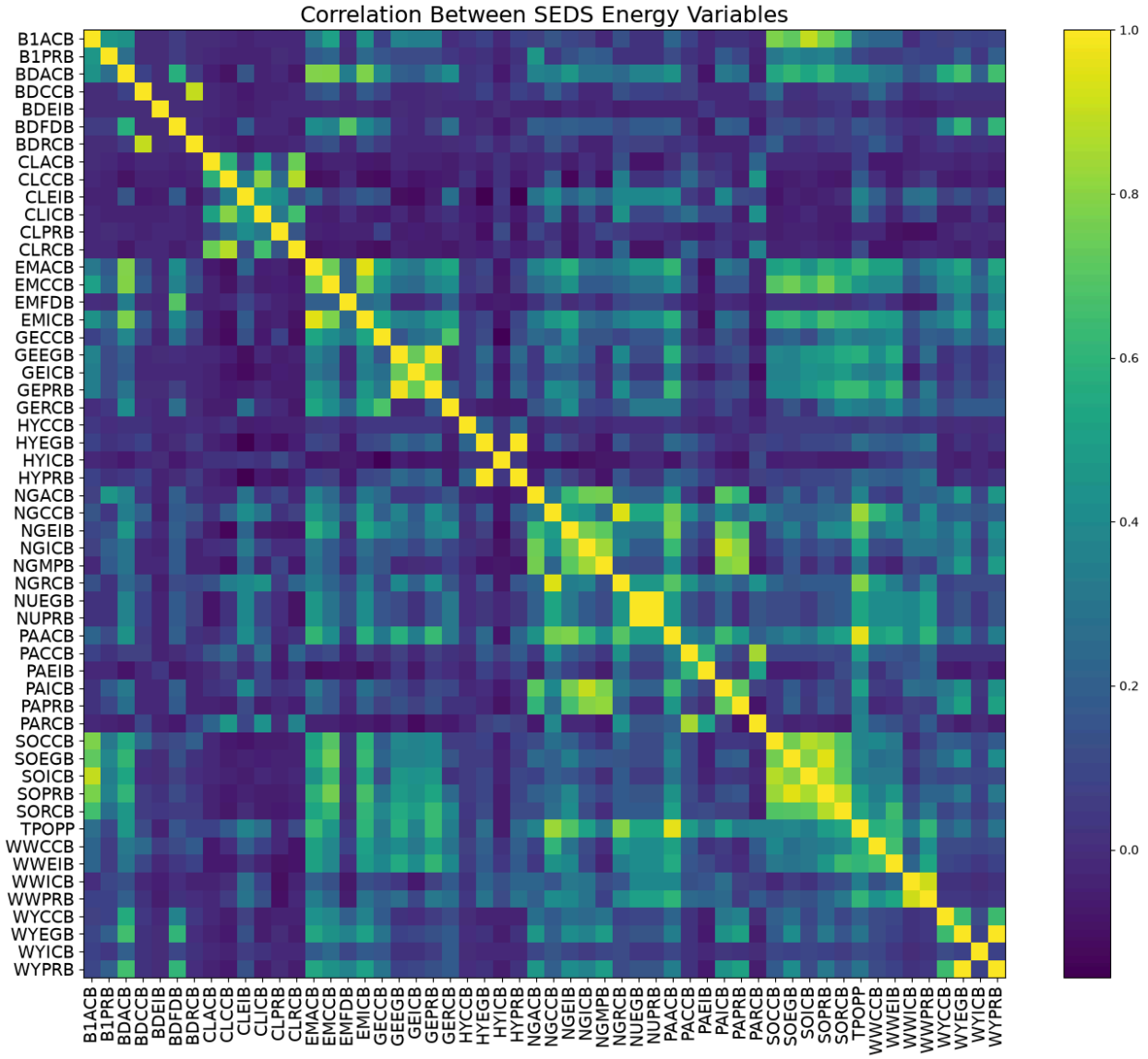
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Appendix

Appendix A:




Correlation Between SEDS Energy Variables
B1ACB
B1PRB
BDACB
BDCCB
BDEIB
BDFDB
BDRCB
CLACB
CLCCB
CLEIB
CLICB
CLPRB
CLRCB
EMACB
EMCCB
EMFDB
EMICB
GECCB
GEEGB
GEICB
GEPRB
GERCB
HYCCB
HYEGB
HYICB
HYPRB
NGACB
NGCCB
NGEIB
NGICB
NGMPB
NGRCB
NUEGB
NUPRB
PAACB
PACCB
PAEIB
PAICB
PAPRB
PARCB
SOCCB
SOEGB
SOICB
SOPRB
SORCB
TPOPP
WWCCB
WWEIB
WWICB
WWPRB
WYCCB
WYEGB
WYICB
WYPRB
1.0
0.8
0.6
0.4
0.2
0.0

Appendix B: Final List of SEDS Features

B1PRB: Renewable diesel | Production
BDACB: Biodiesel | Transportation
BDEIB: Biodiesel | Electric Power
BDFDB: Biodiesel | Production
CLACB: Coal | Transportation
CLCCB: Coal | Commercial
CLEIB: Coal | Electric Power
CLICB: Coal | Industrial
CLPRB: Coal | Production
CLRCB: Coal | Residential
EMACB: Fuel ethanol | Transportation
EMCCB: Fuel ethanol | Commercial
EMFDB: Fuel ethanol | Production
EMICB: Fuel ethanol | Industrial
RECCB: Non-combustible renewables | Commercial
REEIB: Non-combustible renewables | Electric Power
REICB: Non-combustible renewables | Industrial
REPRB: Non-combustible renewables | Production
RERCB: Non-combustible renewables | Residential
NGACB: Natural gas | Transportation
NGCCB: Natural gas | Commercial
NGEIB: Natural gas | Electric Power
NGICB: Natural gas | Industrial
NGMPB: Natural gas | Production
NGRCB: Natural gas | Residential
NUEGB: Nuclear | Electric Power
PAACB: Petroleum | Transportation
PACCB: Petroleum | Commercial
PAEIB: Petroleum | Electric Power
PAICB: Petroleum | Industrial
PAPRB: Petroleum | Production
PARCB: Petroleum | Residential
WWCCB: Wood and waste | Commercial
WWEIB: Wood and waste | Electric Power
WWICB: Wood and waste | Industrial
WWPRB: Wood and waste | Production

Appendix C: Final List of AQI Features

Final AQI features:
Year: Year data was observed
State: US State where the data was collected
Median AQI: Weighted average median AQI
n_counties: Number of counties with data
Days CO: Days with carbon monoxide monitoring
Days NO2: Days with nitrogen dioxide monitoring
Days Ozone: Days with ozone monitoring
Days PM2.5: Days with PM2.5 monitoring
Days PM10: Days with PM10 monitoring

Unsupervised learning output:
Year: Year the data was observed
State: US State where the data was collected
Cluster_id: Cluster the dataset was associated with in the unsupervised learning model